

PROJECT HANDOFF — Network Threat Simulation and Incident Validation Lab

Purpose of this document: paste this entire file into a new Claude chat if the current one gets overloaded. It contains everything needed to continue without re-explaining anything or losing progress.

⚠ **SYNC STATUS — READ THIS FIRST: - GitHub (live, pushed):** <https://github.com/RithikaGowdelli/network-threat-simulation-lab> — currently only has Scenario 1 complete. Everything else below is NOT yet on GitHub. - **This zip / local backup file (network-threat-simulation-lab-BACKUP-not-pushed-to-github.zip):** ahead of GitHub — contains Scenario 1 AND Scenario 3 complete, PLUS a curated /evidence/ folder with 15 essential screenshots, plus this handoff doc. This is the most current version of the work, but it is only saved locally/in this chat's downloads — NOT yet pushed to the live GitHub repo. - **Decision made:** hold off pushing to GitHub until Scenario 2 and Scenario 4 are both finished, then do one single final push covering everything at once. - **If resuming in a new chat:** the zip file is the source of truth for current progress, not the GitHub repo — GitHub is intentionally behind right now. - **Evidence folder:** /evidence/ contains 15 curated screenshots (scenario-01, scenario-03, troubleshooting, github-setup), each with clean filenames and an index at evidence/README.md. Cut down from 60 originally uploaded — only KEY proof points kept. Splunk screenshots deliberately excluded (unresolved work + one had a visible password).

Lab Environment (confirmed working)

Item	Value
Kali Linux IP	192.168.50.10 (labnet, eth0) — resets on every VM reboot , must be re-added manually every time
Metasploitable2 IP	192.168.50.20 (labnet, eth0) — also resets on reboot , same fix needed
Metasploitable2 second adapter	eth1, 192.168.56.20, host-only network, reaches Splunk on Windows host
Splunk	Runs on Windows host (not inside any VM), browser-based. Old account expired, all previously ingested data lost. New 60-day trial signed up — Rithika choosing between Splunk Cloud (recommended: faster, no install) vs Splunk Enterprise (already may have installed this — if so, stick with it rather than switching)
Network mode	VirtualBox Internal Network, "labnet" — fully isolated, no internet inside either VM
Kali OS	Kali Rolling 2026.1, username kali Ubuntu 8.04 (old — lacks

IP fix commands (run after EVERY VM reboot, on the respective machine):

```
sudo ip addr add 192.168.50.10/24 dev eth0 # on Kali
sudo ip addr add 192.168.50.20/24 dev eth0 # on Metasploitable2
sudo ip link set eth0 up
```

Known trap — IP CONFLICT: Kali's IP has previously gotten accidentally set to .20 (same as target) during troubleshooting after a reboot. This causes ping to falsely succeed (Kali pinging itself) while curl/HTTP fails outright with "0 ms, could not connect." If this happens again: check with `ip addr show | grep 192.168.50` on Kali — if it shows .20, run:

```
sudo ip addr del 192.168.50.20/24 dev eth0
sudo ip addr add 192.168.50.10/24 dev eth0
```

Clipboard between Windows host and Kali VM: if copy-paste isn't working, check VirtualBox menu: Devices → Shared Clipboard → Bidirectional. If already set and still not working, just type commands manually rather than troubleshooting further — it's a known low-priority friction point.

Scenario 1 — Network Reconnaissance and Port Scanning: ✓ DONE

- Command: `nmap -sT -sV 192.168.50.20`
- Result: 23 open ports out of 1000 scanned, 15.32 seconds, full output saved in repo
- Wireshark validated: SYN packet burst confirmed via filter `ip.addr==192.168.50.20 && tcp.flags.syn==1 && tcp.flags.ack==0`
- **Confirmed exploited: vsftpd 2.3.4 backdoor (CVE-2011-2523)** on port 21. Trigger: FTP login with username containing `:`, then `nc 192.168.50.20 6200`. Took 4 attempts (3 failed, 1 succeeded) — inconsistent/timing-sensitive trigger. Confirmed root access via `whoami→root, hostname→metasploitable, id→uid=0(root)`.
- Also noted but not tested: port 6667 UnrealIRCd backdoor (CVE-2010-2075), port 1524 pre-existing root bindshell (no auth needed at all)
- Full remediation drafted (3-part: problem/why it matters/fix) for port 21 fix and the undetected-scan finding
- Full evidence written into `scenarios/01-network-reconnaissance/README.md` in the repo — **STATUS: COMPLETE**

Scenario 3 — Suspicious HTTP Path Enumeration: ✓ DONE

- Command: `gobuster dir -u http://192.168.50.20 -w /usr/share/wordlists/dirb/common.txt`
- Result: 4,613 paths tried, found `phpinfo.php`, `phpMyAdmin`, `twiki`, plus expected 403s on `.htaccess/.htpasswd/etc`.
- Wireshark validated: 4,615 displayed HTTP requests matched Gobuster's 4,613 (confirmed via filter `http.request && ip.addr == 192.168.50.20`)
- **phpinfo.php — CONFIRMED information disclosure finding.** Leaked: PHP 5.2.4-2ubuntu5.10, Apache 2.2.8, document root `/var/www/`, config path `/etc/php5/cgi/php.ini`, MySQL client 5.0.51a, Suhosin patch 0.9.6.2. This is the strongest/realest finding in this scenario.
- **phpMyAdmin — CONFIRMED negative.** Tested 4 credential

combos (root/blank, root/root, root/msfadmin, msfadmin/msfadmin), all failed with “Access denied.”

- **Twiki — CONFIRMED negative.** Verified genuinely 2003-era install (revision r1.20 in page footer). Tested CVE-2004-2649 command injection via bin/search with a `echo HELLO_FROM_TWIKI` payload — server returned it as literal text, no execution.
- Full remediation drafted (3-part structure) for all three findings
- Also documented mid-scenario: an IP conflict incident (Kali got assigned .20 by mistake) was diagnosed and fixed — worth keeping in the writeup as a real troubleshooting narrative
- Full evidence written into scenarios/03-http-path-enumeration/README.md in the repo — **STATUS: COMPLETE**

Scenario 2 — Repeated SSH Authentication Failures: IN PROGRESS (just resumed)

- Attack already run in a past session: `hydra -l msfadmin -P /usr/share/wordlists/rockyou.txt ssh://192.168.50.20 → 1,694` failed login entries in `/var/log/auth.log`
- Filtered log saved as `attack_sample3.log` — **saved locally on Rithika's laptop**, ready to upload
- Old Splunk account expired, all previously ingested data lost. **New 60-day trial just signed up and confirmed ready as of this handoff.**
- SPL query logic known (rex to extract IP, 5-minute bins, threshold >5 failures/IP) but exact query text was never saved — needs to be rebuilt from scratch in the new Splunk instance
- Previously also unresolved: Splunk alert wouldn't save, error “enable at least one action” — needs to be resolved fresh in the new trial
- **Immediate next steps when resuming:**
 1. Confirm both VM IPs are set (192.168.50.10 Kali, 192.168.50.20 Metasploitable2) after reboot
 2. Log into new Splunk instance (Cloud or Enterprise, browser-based on Windows host)
 3. Upload `attack_sample3.log` as a new data source
 4. Set sourcetype to `linux_secure`
 5. Confirm the data landed with a basic search before touching detection logic
 6. Rebuild the SPL query (rex extraction, 5-min bins, >5 threshold) from scratch, test it, screenshot everything this time — no screenshots exist from the original run
 7. Fix the alert-save error if it recurs
 8. Write full Scenario 2 README section (same structure as Scenario 1 and 3) using real evidence

Scenario 4 — Controlled Unexpected Outbound Connection: ☐ NOT STARTED

- Template exists in repo, no work done yet
- Needs a decision first: reverse shell trigger (netcat listener on Kali, triggered shell on Metasploitable2 connecting back) vs. simulated periodic beacon — Rithika needs to choose before starting
- Will follow the same evidence pattern: Wireshark capture, confirm via listener output, document, remediate

GitHub Repo

Status: pushed once already, one commit behind as of this handoff. Live at: <https://github.com/RithikaGowdelli/network-threat-simulation-lab>

Local repo on Rithika's Windows laptop: D:\network-threat-simulation-lab, connected to the above remote, branch main.

Decision made: hold off on pushing Scenario 3's update until Scenario 2 and 4 are both done — do ONE final push at the end covering everything, rather than pushing after every scenario. This avoids repeated file-sync friction between Claude's sandbox copy and Rithika's local Windows copy.

When ready for the final push: Claude will provide a fresh zip of the complete repo (with full git history) to download, extract over the existing local folder, then:

```
cd D:\network-threat-simulation-lab
git remote add origin https://github.com/RithikaGowdelli/network-
threat-simulation-lab.git
git push -u origin main
```

Should be a clean fast-forward push since local commits build directly on what's already on GitHub.

Resume Content (drafted, DO NOT send to employers until Scenario 2 & 4 are done)

Three portfolio project bullet sets were drafted together (in order): 1) SOC Alert Monitoring & Detection Engineering Lab (Splunk/Sysmon), 2) this project (Network Threat Simulation & Incident Validation Lab), 3) Microsoft Sentinel Cloud Threat Hunting Lab. Only project 2's bullets are reproduced here since it's the one being actively worked on:

Built an isolated cybersecurity lab using Kali Linux, Metasploitable, Splunk, Wireshark, and Nmap to simulate and investigate 4 controlled attack scenarios spanning reconnaissance, authentication attacks, web enumeration, and network-based threat activity. Scanned 1,000 TCP ports against an isolated vulnerable target using Nmap, identifying 23 exposed services in 15.32 seconds and mapping attack surface exposure through service enumeration. Simulated 1,694 failed SSH authentication events using Hydra to emulate brute-force activity, generating telemetry for SIEM detection development and incident investigation. Performed web content enumeration using Gobuster across 4,613 directory paths, cross-validating 4,615 corresponding requests through Wireshark packet analysis to confirm application activity at the network level. Correlated TCP/IP, DNS, HTTP, and SSH traffic with Splunk security logs to validate alerts, reconstruct attack timelines, and extract 15+ Indicators of Compromise (IOCs). **(PLACEHOLDER — update after Scenario 2/4 complete, real IOC count not yet confirmed.)** Developed 5+ SPL detection searches for reconnaissance and authentication-based attacks, mapped validated activity to MITRE ATT&CK, and produced 4 incident investigation reports with remediation recommendations. **(PLACEHOLDER — update after Scenario 2/4 complete, SPL query count and incident report count not yet real.)**

Do not send this resume version to any employer until the two placeholder bullets have real numbers.

Standing Rules for This Project (apply throughout)

- Never invent evidence, metrics, screenshots, or field names — always ask for real command output

- Every scenario needs: objective, prerequisites, procedure, logs/evidence, Wireshark filters, Splunk workflow, SPL queries, investigation questions, IOCs, timeline, true/false positive assessment, MITRE mapping, remediation, screenshot checklist, status
 - Remediation recommendations are drafted by Rithika first using a 3-part structure (Problem / Why it matters / Fix), then reviewed/tightened by Claude — never written from scratch by Claude
 - Lessons-learned section must be written by Rithika in her own words, not by Claude
 - All testing confined to the isolated labnet — no real/public systems ever targeted
 - Commit messages should be specific and incremental, matching real work done that session, not generic
 - When something is tested and fails (negative result), that's documented as a legitimate finding, not treated as an error to hide or keep retrying indefinitely
 - All commands run FROM Kali, aimed AT Metasploitable2 — Metasploitable2 is only logged into directly to check its own internal state (processes, firewall rules, IP config), never to launch attacks against itself
-

What Happens Next (pick up here)

1. **Currently resuming Scenario 2** — Splunk trial confirmed ready. Do the 8 steps listed under Scenario 2 above.
2. After Scenario 2: do Scenario 4 (decide reverse shell vs. beacon approach first)
3. Do ONE final GitHub push covering Scenario 2 + 4 + any final touches
4. Build the final consolidated document with all screenshots embedded
5. Write lessons-learned section (Rithika's own words — Claude prompts with questions but does not write it)
6. Finalize architecture doc and detection-engineering docs once all scenarios are locked in
7. Update resume bullets with real final numbers once Scenario 2 & 4 are done — remove placeholder markers